Note on A course in Arithmetic by J.P. Serre

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Contents

1 Chapter 1

This is a list of proofs of nontrivial results in Chapter 1 of *A course in Arithmetic* by J.P. Serre.

Let K be a finite field with q elements.

Theorem 1.1

Let u be an integer ≥ 0. The sum

$$S(X^u) = \sum_{x \in K} x^u = \begin{cases} -1 & \text{if } u > 0 \text{ and } (q-1) \mid u \\ 0 & \text{otherwise} \end{cases}$$

$u = 0$ is subtle here. We define $x^0 = 1$ for all $x \in K$ (this definition is nice when we work with polynomials because we can write 1 as x^0 to maintain symmetry), so $S(X^0) = q = 0$.

The proof is easy. Use primitive roots.
The variant proof: $x \mapsto x^u$ is a group homomorphism from $K^\times$ to itself. The kernel is all x such that x is a root of $x^{\gcd(u,q-1)} - 1$. So the image has size $\frac{q-1}{\gcd(u,q-1)}$, which implies the image is just the $d = \frac{q-1}{\gcd(u,q-1)}$ roots of unity. So the sum is 0 unless $d = 1$ (sum of d-th roots of unity is 0 if $d \geq 2$ s coprime to p).

Theorem 1.2 (Chevalley-Warning Theorem)

Let $P_1, \ldots, P_r$ be polynomials in n variables over a finite field K of characteristic p. If the sum of their degrees is less than n, then the number of common solutions to the equations $P_1 = \cdots = P_r = 0$ is divisible by p.

Proof. This uses a common trick in counting roots over finite fields. Consider the polynomial

$$Q = \prod_{i=1}^{r} (1 - P_i^{q-1})$$

The number of common solutions is equal to $\sum_{x \in K^n} Q(x)$ since if $P_i(x) = 0$ for all i, then $Q(x) = 1$, otherwise $Q(x) = 0$.

Expand this. For each monomial in the expansion $x_1^{u_1} \cdots x_n^{u_n}$, since $\sum \deg(P_i)(q-1) < n(q-1)$, we have u_j is not divisible less than $q-1$ for some j. So by the previous theorem, the sum over K^n of this monomial is 0. (that's why we define $x^0 = 1$ above). So the number of common solutions is 0 mod p. $\square$

Corollary 1.3

All quadratic forms in at least 3 variables over a finite field have nontrivial zeros.

Theorem 1.4

$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$

Proof. Let α be the primitve 8th root of unity α in $\overline{\mathbb{F}_p}$. Then, $y^2 = (\alpha+\alpha^{-1})^2 = 2$. If $p \equiv \pm 1 \mod 8$, then $y^p = y$, so 2 is a square in $\mathbb{F}_p$. If $p \equiv \pm 3 \mod 8$, then $y^p = -y$, so 2 is not a square in $\mathbb{F}_p$. $\square$

Theorem 1.5 (Law of Quadratic Reciprocity)

For two odd primes p and q, we have

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{(p-1)(q-1)}{4}}$$

The key is to consider the Gauss sum: Let ζ be a primitive p-th root of unity in $\overline{\mathbb{F}_p}$, then define

$$y = \sum_{x \in \mathbb{F}_p} \left(\frac{x}{p}\right) \zeta^x$$

Then we have $y^2 = \left(\frac{-1}{p}\right) p = (-1)^{\frac{p-1}{2}} p$ and $y^{q-1} = \left(\frac{q}{p}\right)$. Combining these two gives the result.

2 Meeting 1 on Jan. 28 2026

This is a list of nontrivial results in the first meeting of the course.

Theorem 2.1

The finite subgroup of a field's multiplicative group is cyclic.

This can be easily proven using the structure theorem of finite abelian groups and the fundamental theorem of Algebra.

Definition 2.2. For any prime p or $p = \infty$, we define the

$$\|x\|_p = \begin{cases} p^{-v_p(x)} & p \text{ is a prime} \\ \|x\| & p = \infty \end{cases}$$

Theorem 2.3

For any rational number r, $\prod_p \|r\|_p = 1$

This is trivial from the definition of v_p.

We can in general define a norm on a number field K as follows:

Definition 2.4. $\|\cdot\| : K \to \mathbb{R}_{\geq 0}$ is a norm if

1. $\|x\| = 0$ iff $x = 0$
2. $\|xy\| = \|x\| \|y\|$
3. $\|x + y\| \leq \|x\| + \|y\|$

Corollary 2.5

$\|1\| = 1$ and $\|-1\| = 1$

Definition 2.6. Norm N_1 and N_2 are equivalent if there exists a real number c such that $N_2(x) = N_1(x)^a$ for all x.

Theorem 2.7 (Ostrowski's Theorem)

Any non-trivial norm on $\mathbb{Q}$ is equivalent to either the usual absolute value $\|\cdot\|_\infty$ or to $\|\cdot\|_p$ for some prime p.

Proof. □

Theorem 2.8 (Hensel's Lemma)

Proof. □

Theorem 2.9 (Hasse principle)

Example 2.10

$f(x, y, z) = 5x^2 + 7y^2 - 13z^2$. We want to find a nontrivial $f(x, y, z) = 0$ has non-trivial solutions in $\mathbb{Q}$ using Hensel's Lemma.

Consider this curve in $\mathbb{P}^3$, is there any other solution?

Use parameterization of this curve.

Theorem 2.11

If $n \geq 3$, $x^n + y^n = z^n$ has no non-trivial solutions in $\mathbb{C}[x]$.

Proof. Assume f, gh are non-constant polynomials such that $f^n + g^n = h^n$. We further assume they $\max \deg(f), \deg(g), \deg(h)$ is minimal among all such triples. Therefore, $\gcd(f, g) = \gcd(g, h) = \gcd(h, f) = 1$. (otherwise, we can divide the common factor to get a smaller degree triple).

$f^n = h^n - g^n = \prod(h - \zeta^i g)$ where ζ is a primitive n-th root of unity. Notice that $\gcd(h - \zeta^i g, h - \zeta^j g) = \gcd(g, h)$ for $i \neq j$. So all $h - \zeta^i g$ are coprime to each other. Thus, $h - g = a^n$, $h - \zeta g = b^n$, $z - \zeta^2 g = c^n$ for some non-constant polynomials $a, b, c \in \mathbb{C}[x]$. We can easily find $p, q, r \in \mathbb{C}$ such that $(pa)^n + (qb)^n + (rc)^n = 0$. This contradicts the minimality of our choice of f, g, h. $\square$

3 Chapter 2

3.1 $\mathbb{Z}_p$ and $\mathbb{Q}_p$

Let $A_n = \mathbb{Z}/p^n\mathbb{Z}$. There is a natural projection map $\phi_n : A_n \to A_{n-1}$.

Definition 3.1. We define $\mathbb{Z}_p$ as the limit (in category of rings) of

$$\cdots \xrightarrow{\phi_{n+1}} A_n \xrightarrow{\phi_n} A_{n-1} \xrightarrow{\phi_{n-1}} \cdots \xrightarrow{\phi_2} A_1$$

By the definition, any $x \in \mathbb{Z}_p$ can be represented as a seqeunce $(\ldots, x_2, x_1)$ where $x_n \in A_n$ and $\phi_n(x_n) = x_{n-1}$. Alternatively, we can represent x as an infinite sum

$$x = a_0 + a_1 p + a_2 p^2 + \cdots$$

where $a_i \in \{0, 1, \ldots, p-1\}$.

We can also define the topology on $\mathbb{Z}_p$ as the induced topology from the product topology on $\prod A_n$ where A_n is given the discrete topology.

Theorem 3.2

$\mathbb{Z}_p$ is compact.

Proof. We will need a lemma from topology.

Lemma 3.3

Closed subset of a compact space is compact.

Proof. If K is compact and $F \subseteq K$ is closed, then for any open covering of F, we can add $K \setminus F$ to get an open covering of K. So there is a finite subcovering, removing $K \setminus F$ gives a finite subcovering of F. $\square$

Since $\prod A_n$ is compact by Tychonoff's theorem, it suffices to show $\mathbb{Z}_p$ is closed in $\prod A_n$. It suffices show that its complement is open. For any $y \notin \mathbb{Z}_p$, there exists some j such that $\phi_j(y_j) \neq y_{j-1}$. There exists an open set $U = \prod U_n$ where $U_n = A_n$ for $n \neq j, j-1$, $U_j = \{y_j\}$ and $U_{j-1} = \{y_{j-1}\}$. It's clear that U is an open neighborhood of y and $U \cap \mathbb{Z}_p = \emptyset$. So $\mathbb{Z}_p$ is closed. $\square$

Theorem 3.4

$$0 \to A_m \xrightarrow{\times p^n} A_{m+n} \xrightarrow{\phi_{m+n,n}} A_n \to 0$$

is an exact seqeunce of abelian groups.

This is easy to check. The first injective map is moving the the last m digits $\times p^n$ to the left by n places and fill the last n digits with zero, the second map is just truncating the last n digits. We will use this in $\mathbb{Z}_p$.

Theorem 3.5

$$0 \to \mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p \to A_n \to 0$$

is an exact seqeunce of abelian groups.

Proof. First, we need to show $\mathbb{Z}_p \xrightarrow{\times p^n} \mathbb{Z}_p$ is injective. Notice that if $\phi_{m+n,n}(x_{m+n}) = x_n$, then $p^m x_n$ and $p^m x_{m+n}$ are the same in A_{m+n}. Therefore, we conclude that multiplying by p^n is just adding n zeros to the end, that is sending $(\ldots, x_2, x_1)$ to $(\ldots, p^n x_2, p^n x_1, 0, \ldots, 0)$. Let j be the smallest integer such that the j-th digit of x is nonzero (not divisible by p^j), then the $j+n$-th digit of $\times p^n x$ is also nonzero since it's not divisible by p^{j+n}, so the map is injective.

The image of $\times p^n$ is just $p^n\mathbb{Z}_p$, which is exactly the kernel of the projection map to A_n since the last n digits are all zero. $\square$

Theorem 3.6

Two propositions:

1. For an element $x \in \mathbb{Z}_p$ to be invertible, it is equivalent to say that it's not divisible by p, i.e., the first digit a_0 in its expansion is nonzero.

2. Let U be the group of units in $\mathbb{Z}_p$. Then we have the decomposition. For any $x \in \mathbb{Z}_p$, we can write

$$x = p^n u$$

where $n \geq 0$ and $u \in U$ in a unique way.

For part 1, if x_1 is invertible, a_i are corpime to p. By Bezout's lemma, there exists $a_i, b_i \in \mathbb{Z}$ such that $a_i x_i + b_i p^i = 1$. Let $a = (\ldots, a_2, a_1)$, this is clearly the inverse of x and we can check it's indeed in $\mathbb{Z}_p$.

Other parts are easy.

Proposition 3.7

As a ring, we can check that $\mathbb{Z}_p$ is an integral domain. In fact, it's a local ring with maximal ideal $p\mathbb{Z}_p$.

Definition 3.8. We define $v_p(x) = n$ if $x = p^n u$ where u is a unit in $\mathbb{Z}_p$. We define $v_p(0) = +\infty$.

Proposition 3.9

$v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x+y) \geq \inf\{v_p(x), v_p(y)\}$ with equality if $v_p(x) \neq v_p(y)$.

This allows us to define a metric on $\mathbb{Z}_p$ as follows:

Definition 3.10. $d(x, y) = p^{-v(x-y)}$

A notbale property is that $d(x, y) = d(x - z, y - z)$ for any $x, y, z \in \mathbb{Z}_p$. So the metric is translation invariant. In particular, we have $d(x, y) = d(0, y - x)$.

Theorem 3.11

The metric space $(\mathbb{Z}_p, d)$ is homomorphic to the topological space $\mathbb{Z}_p$ defined as the subspace of $\prod A_n$ with the product topology.

Proof. It suffices to show that the identity map from $(\mathbb{Z}_p, d)$ to $\mathbb{Z}_p$ is continuous in both directions.

For any open set $U \subseteq \mathbb{Z}_p$,

$$U = \mathbb{Z}_p \cap \left(\bigcup_{i \in I} (\prod U_{n,i}) \right) = \bigcup_{i \in I} \left((\prod U_{n,i} \cap \mathbb{Z}_p) \right)$$

where $U_{n,i} = A_n$ for all but finitely many n.

What does $T_i = \prod U_{n,i} \cap \mathbb{Z}_p$ looks like? Let N be the largest integer such that $U_{N,i} \neq A_N$. Then, T_i is the set of x such that the first N digits of x are fixed and the rest can be anything possible, that is $x = (\ldots, a_N, \ldots, a_1)$ where $a_1, \ldots, a_N$ are fixed. The preimage of T_i in $(\mathbb{Z}_p, d)$ is precisely the open ball centered at x with radius p^{-N}, that is $x + p^N \mathbb{Z}_p$.

Conversely, what does any open ball in $(\mathbb{Z}_p, d)$ look like in $\mathbb{Z}_p$? First, the open ball of 0 is just ideals $p^n \mathbb{Z}_p$. For other point, if the radius of the open abll is p^{-N}, for some $N \in \mathbb{Z}$. $d(x, y) = d(0, x - y) < p^N$ means $v_p(x - y) > N$, which means the first N digits of x and y are the same. So the open balls are the cosets of $p^n \mathbb{Z}_p$. So the preimage of this open ball is the set of y such that the first N digits are fixed, which is clearly open in $\mathbb{Z}_p$. □

Remark 3.12. Any open ball is also closed in $\mathbb{Z}_p$, since the image of $d(0, x)$ is discrete in $\mathbb{R}$. So $\mathbb{Z}_p$ is totally disconnected.

Remark 3.13. $\mathbb{Z}_p$ is complete under this metric. This follows from the fact that $\mathbb{Z}_p$ is compact.

Remark 3.14. $\mathbb{Z}$ is dense in $\mathbb{Z}_p$. Just find an integer with the same first n digits as the element in $\mathbb{Z}_p$ you want to approximate.

Definition 3.15. We define $\mathbb{Q}_p$ as the field of fractions of $\mathbb{Z}_p$. Equivalently, it's the localization of $\mathbb{Z}_p$ at $U = 1, p, p^2, \ldots$.

Definition 3.16. For any $x \in \mathbb{Q}_p$, we can write $x = p^n u$ where $n \in \mathbb{Z}$ and $u \in U$. We define $v_p(x) = n$ and $v_p(0) = +\infty$.

Proposition 3.17

propositions about $\mathbb{Q}_p$:

1. $x \in \mathbb{Q}_p$ is in $\mathbb{Z}_p$ iff $v_p(x) \geq 0$.
2. The metric $d(x, y) = p^{-v_p(x-y)}$ makes $\mathbb{Q}_p$ a topological field.
3. $\mathbb{Z}_p$ is a clopen subset of $\mathbb{Q}_p$.
4. $\mathbb{Q}_p$ is locally compact, that is for any $x \in \mathbb{Q}_p$, there exists open set U and compact set K such that $x \in U \subseteq K$.
5. $\mathbb{Q}$ is dense in $\mathbb{Q}_p$.

For any open neighborhood of $x \in \mathbb{Q}_p$, it looks like $x + p^n\mathbb{Z}_p$ for some n (in fact clopen). Notice that translation and scaling are homeomorphisms, so any neighborhood of x is compact. Others are easy to check.

Proposition 3.18

The metric d is an ultrametric, that is it satisfies the strong triangle inequality

$$d(x, z) \leq \max\{d(x, y), d(y, z)\}$$

for any $x, y, z \in \mathbb{Q}_p$.

As a result, $d(u_m, u_n) \leq \sup_{k \geq n} d(u_k, u_{k+1})$ for any sequence $\{u_n\}$ in $\mathbb{Q}_p$ and any $m > n$. Therefore, a sequence $\{u_n\}$ is Cauchy iff $d(u_n, u_{n+1}) \to 0$ as $n \to \infty$ iff $\lim_{n\to\infty} u_{n+1} - u_n = 0$.